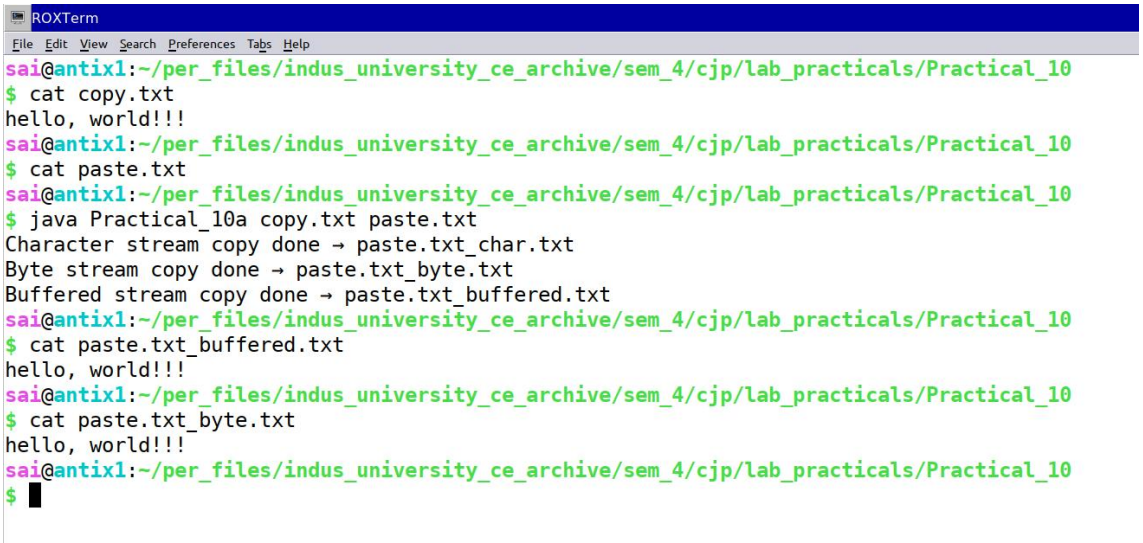
**Aim(10.1): Write a program that takes two files names (source and and destination) as command line argument. Copy source file's content to destination file. Use character stream class. Also do same using byte stream and buffer stream.**

```
import java.io.*;

public class Practical10_1 {

    // 1. Character Stream — FileReader / FileWriter
    static void copyCharacterStream(String src, String dst) throws IOException {
        try (FileReader reader = new FileReader(src);
             FileWriter writer = new FileWriter(dst + "_char.txt")) {
            int ch;
            while ((ch = reader.read()) != -1) {
                writer.write(ch);
            }
        }
        System.out.println("Character stream copy done → " + dst + "_char.txt");
    }

    // 2. Byte Stream — FileInputStream / FileOutputStream
    static void copyByteStream(String src, String dst) throws IOException {
        try (FileInputStream fis = new FileInputStream(src);
             FileOutputStream fos = new FileOutputStream(dst + "_byte.txt")) {
            int byteData;
            while ((byteData = fis.read()) != -1) {
                fos.write(byteData);
            }
        }
        System.out.println("Byte stream copy done → " + dst + "_byte.txt");
    }

    // 3. Buffered Stream — BufferedReader / BufferedWriter
    static void copyBufferedStream(String src, String dst) throws IOException {
        try (BufferedReader br = new BufferedReader(new FileReader(src));
             BufferedWriter bw = new BufferedWriter(new FileWriter(dst +
"_buffered.txt"))) {
            String line;
            while ((line = br.readLine()) != null) {
                bw.write(line);
                bw.newLine();
            }
        }
        System.out.println("Buffered stream copy done → " + dst + "_buffered.txt");
    }

    public static void main(String[] args) {
```

```
        if (args.length < 2) {
            System.out.println("Usage: java Practical10_1 <source> <destination>");
            return;
        }
        String source      = args[0];
        String destination = args[1];

        try {
            copyCharacterStream(source, destination);
            copyByteStream(source, destination);
            copyBufferedStream(source, destination);
        } catch (IOException e) {
            System.out.println("Error: " + e.getMessage());
        }
    }
}
```

**Output(10.1):**



**Aim(10.2): Write a program which generates random integers and stores them in a file named "rand.dat". The program then reads the integers from the file and displays on the screen.**

```
import java.io.*;
import java.util.Random;

public class Practical10_2 {
    private static final String FILE_NAME = "rand.dat";
    private static final int COUNT = 20;

    public static void main(String[] args) {
        Random random = new Random();

        // STEP 1: Write random integers to file
        System.out.println("Writing " + COUNT + " random integers to " + FILE_NAME +
":");
        try (DataOutputStream dos = new DataOutputStream(
                new FileOutputStream(FILE_NAME))) {
            for (int i = 0; i < COUNT; i++) {
                int num = random.nextInt(1000);
                dos.writeInt(num);
                System.out.print(num + " ");
            }
            System.out.println("\nWrite complete.");
        } catch (IOException e) {
            System.out.println("Write error: " + e.getMessage());
            return;
        }

        // STEP 2: Read integers back from file and display
        System.out.println("\nReading integers from " + FILE_NAME + ":");
        try (DataInputStream dis = new DataInputStream(
                new FileInputStream(FILE_NAME))) {
            for (int i = 0; i < COUNT; i++) {
                System.out.print(dis.readInt() + " ");
            }
            System.out.println("\nRead complete.");
        } catch (IOException e) {
            System.out.println("Read error: " + e.getMessage());
        }
    }
}
```

**Output(10.2):**

```
File  Edit  View  Search  Preferences  Tabs  Help
sai@antix1:~/per_files/indus_university_ce_archive/sem_4/cjp/lab_practicals/Practical_10
$ java Practical_10b
Writing 20 random integers to rand.dat:
39 639 425 838 988 682 669 336 138 140 441 856 486 213 378 797 699 537 865 409
Write complete.

Reading integers from rand.dat:
39 639 425 838 988 682 669 336 138 140 441 856 486 213 378 797 699 537 865 409
Read complete.
sai@antix1:~/per_files/indus_university_ce_archive/sem_4/cjp/lab_practicals/Practical_10
$
```